

Zadanie 4 - Robert Mesek

bitmap_h.m

```
function
bitmap_h(filename,elementsx,elementsy,maxerror,max_refinement_level,color_edges_black,mode)
% [...]
RR = XX(:, :, 1); %Red color [0,255]
GG = XX(:, :, 2); %Green color [0,255]
BB = XX(:, :, 3); %Blue color [0,255]
RR_out = zeros(size(RR), 'uint8');
GG_out = zeros(size(GG), 'uint8');
BB_out = zeros(size(BB), 'uint8');

if strcmp(mode, 'R')
    GG = zeros(size(GG));
    BB = zeros(size(BB));
end
if strcmp(mode, 'G')
    RR = zeros(size(RR));
    BB = zeros(size(BB));
end
if strcmp(mode, 'B')
    RR = zeros(size(RR));
    GG = zeros(size(GG));
end
% [...]
redo_error_test = true;
refinemenet_level = 0;

% interpolate all active elements - recreate bitmap red green and blue compoments
for i=1:total_elements
    if (elements(i).active)
        interpolate_elem(i,color_edges_black);
    end
end

% recreate bitmap from red, green and blue compoments
RGB=XX;
RGB(:, :, 1) = RR_out;
RGB(:, :, 2) = GG_out;
RGB(:, :, 3) = BB_out;

% display image
% imshow(RGB);

% save image
filename_out = sprintf('out/%s_%s_%02d_%s', mode, string(color_edges_black),
refinemenet_level, filename);
```

```

imwrite(RGB, filename_out);
% [...]
RGB=XX;
RGB(:, :, 1) = RR_out;
RGB(:, :, 2) = GG_out;
RGB(:, :, 3) = BB_out;

% display image
% imshow(RGB);

% save image
filename_out = sprintf('out/%s_%s_%02d_%s', mode, string(color_edges_black),
refinemenet_level, filename);
imwrite(RGB, filename_out);

end
% [...]
function interpolate_elem(element,color_edges_black)
% [...]
for xx=0:width
    for yy=0:hight
        [r,g,b] = inpoint(xx,yy,width,hight,elem);
        RR_out(dlx+xx,dly+yy) = r;
        GG_out(dlx+xx,dly+yy) = g;
        BB_out(dlx+xx,dly+yy) = b;
    end
end

% create black edges on element if requested
if (color_edges_black)
    for xx=0:width
        RR_out(dlx+xx,dly) = 0;
        GG_out(dlx+xx,dly) = 0;
        BB_out(dlx+xx,dly) = 0;

        RR_out(dlx+xx,dly+hight) = 0;
        GG_out(dlx+xx,dly+hight) = 0;
        BB_out(dlx+xx,dly+hight) = 0;
    end

    for yy=0:hight
        RR_out(dlx,dly+yy) = 0;
        GG_out(dlx,dly+yy) = 0;
        BB_out(dlx,dly+yy) = 0;

        RR_out(dlx+width,dly+yy) = 0;
        GG_out(dlx+width,dly+yy) = 0;
        BB_out(dlx+width,dly+yy) = 0;
    end
end
end
% [...]

```

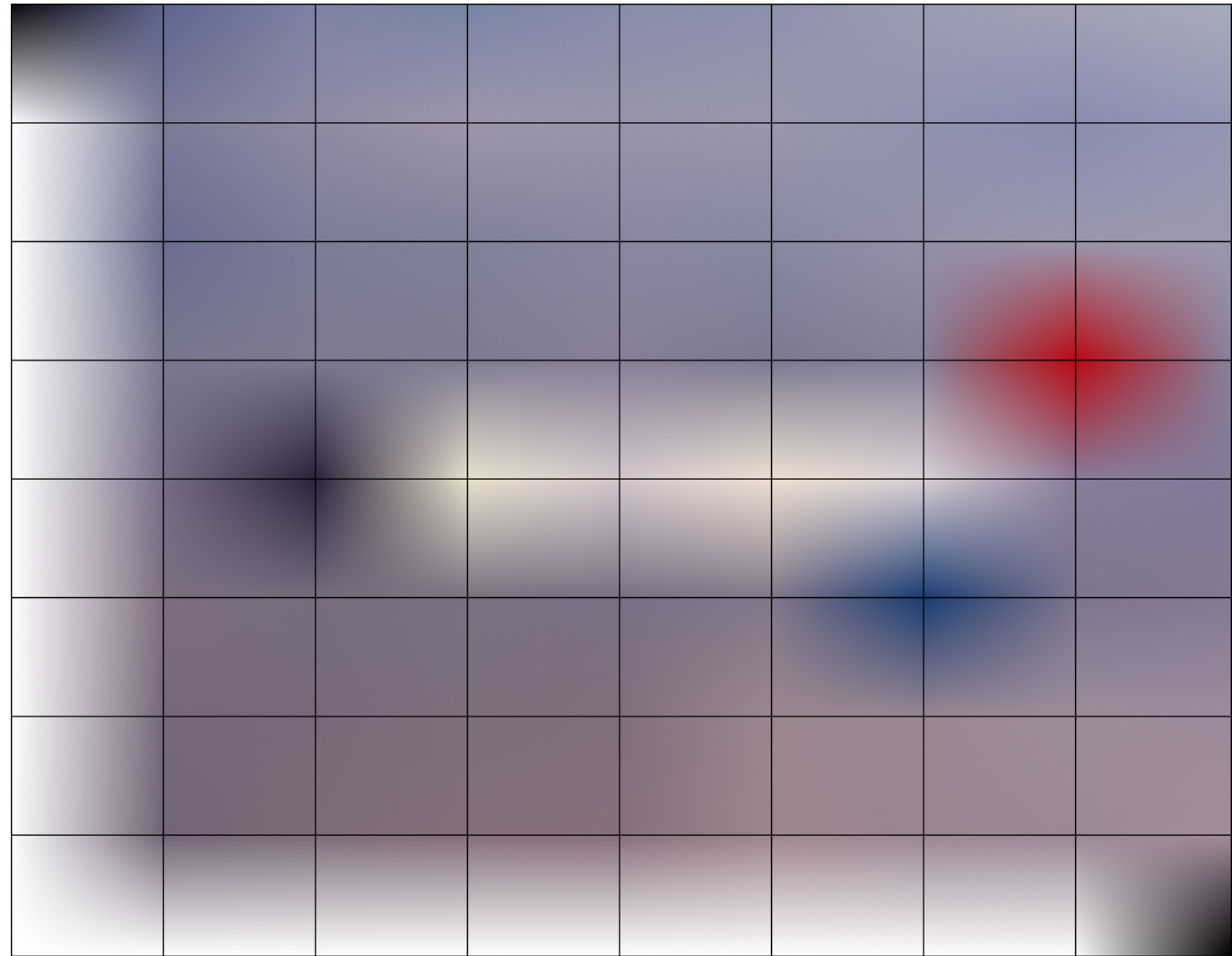
run.m

```
bitmap_h("yf16.bmp",8,8,2,8,true,'RGB');  
bitmap_h("yf16.bmp",8,8,2,8,false,'RGB');  
bitmap_h("yf16.bmp",8,8,2,8,true,'R');  
bitmap_h("yf16.bmp",8,8,2,8,false,'R');  
bitmap_h("yf16.bmp",8,8,2,8,true,'G');  
bitmap_h("yf16.bmp",8,8,2,8,false,'G');  
bitmap_h("yf16.bmp",8,8,2,8,true,'B');  
bitmap_h("yf16.bmp",8,8,2,8,false,'B');
```

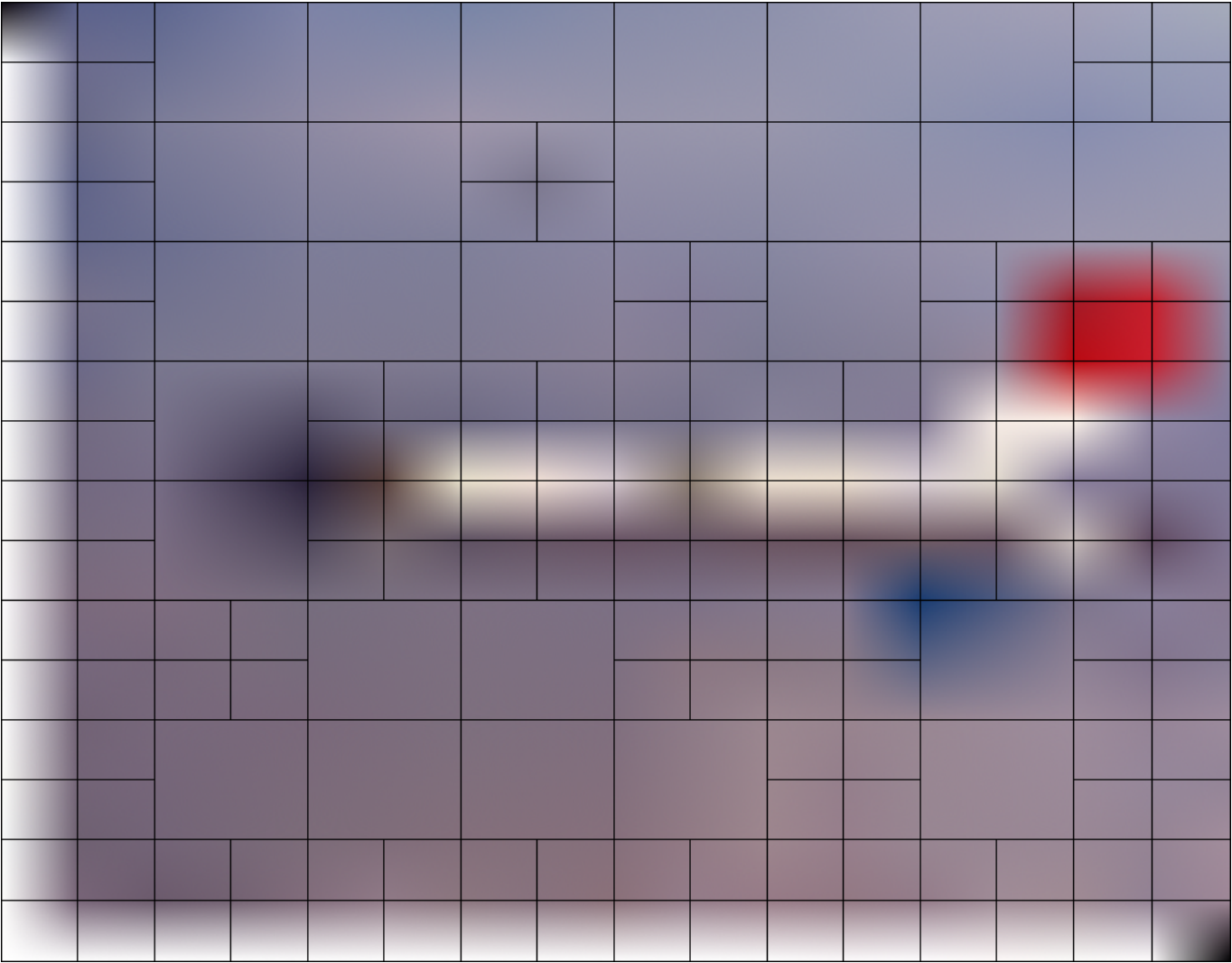
yf16.bmp



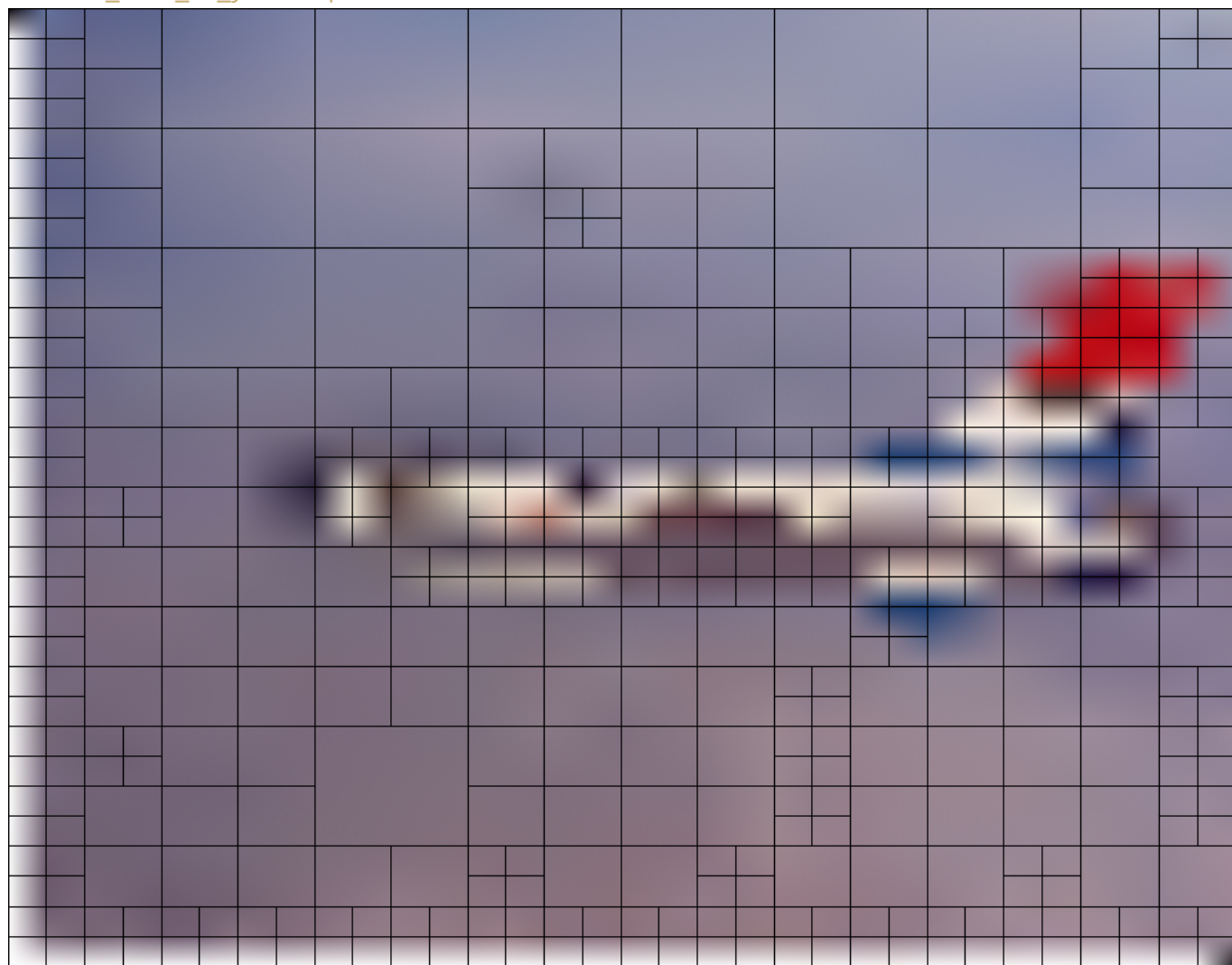
The figure displays a heatmap of a 2D function on an 8x8 grid. The color scale ranges from dark blue (low values) to dark red (high values). The function exhibits a strong positive correlation in the top-right corner, indicated by the red color, and a strong negative correlation in the bottom-left corner, indicated by the dark blue color. The central region shows a yellowish-green color, suggesting intermediate values. The grid lines are thin and black, separating the 64 individual cells of the 8x8 matrix.



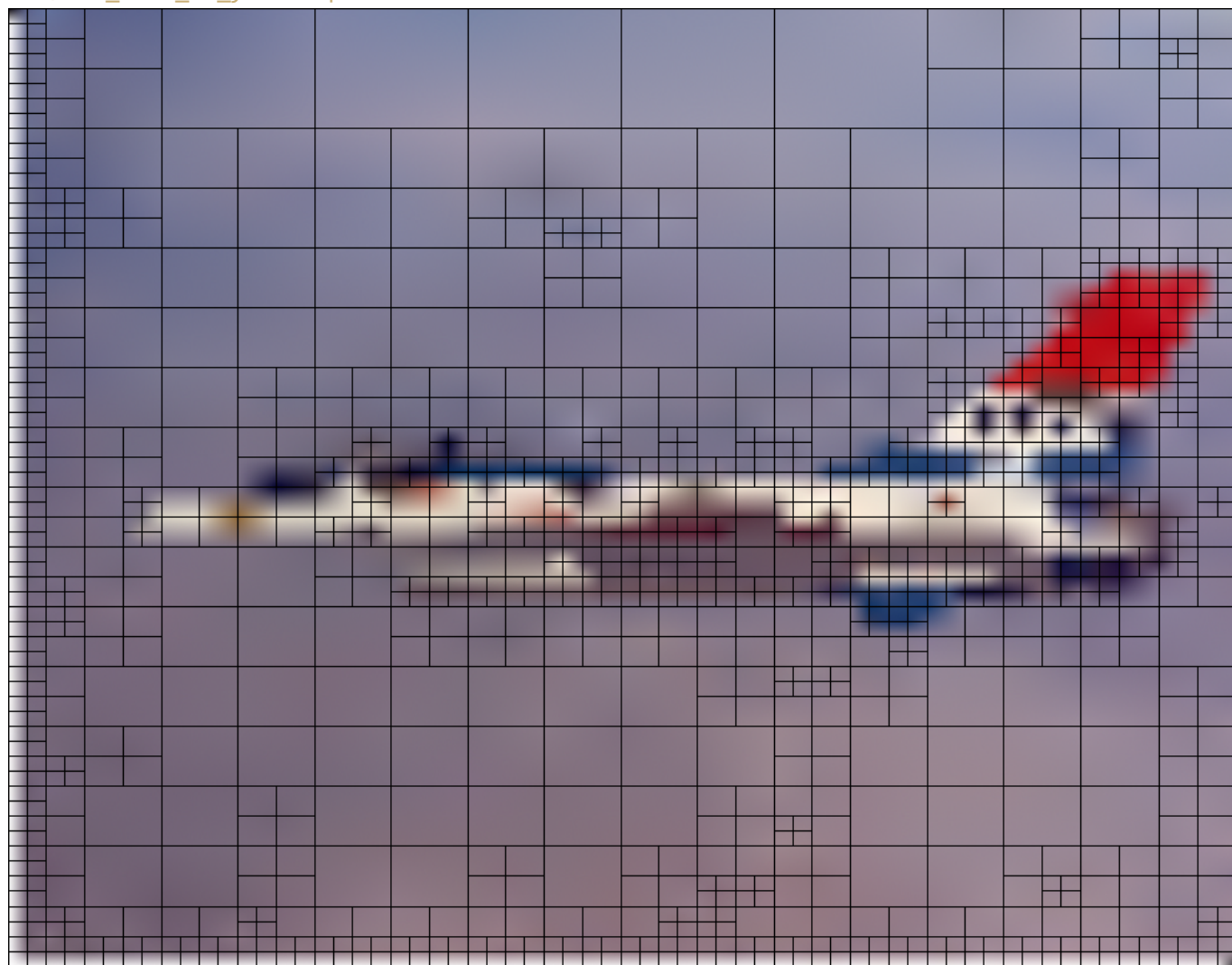
out/RGB_true_01_yf16.bmp



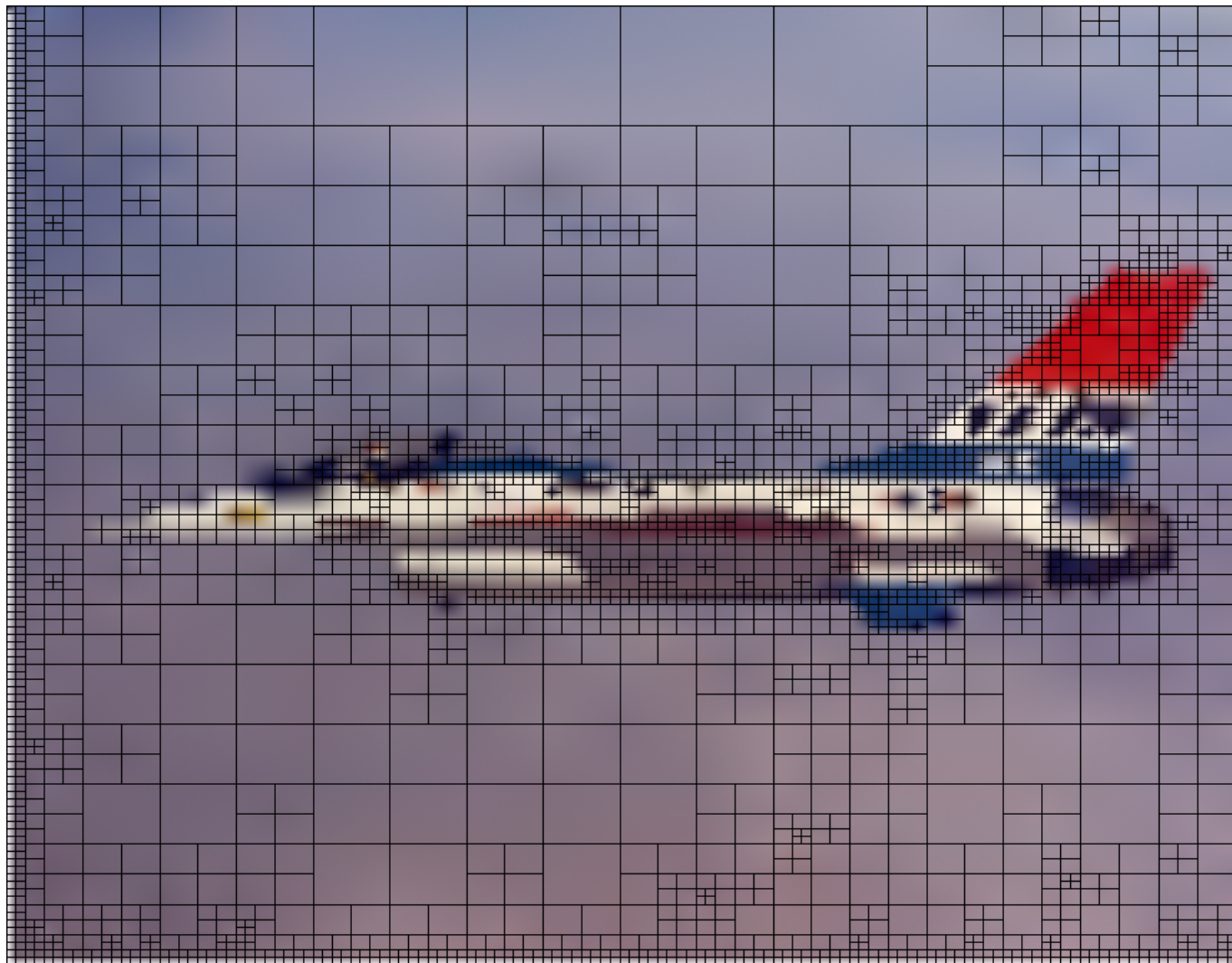
out/RGB_true_02_yf16.bmp



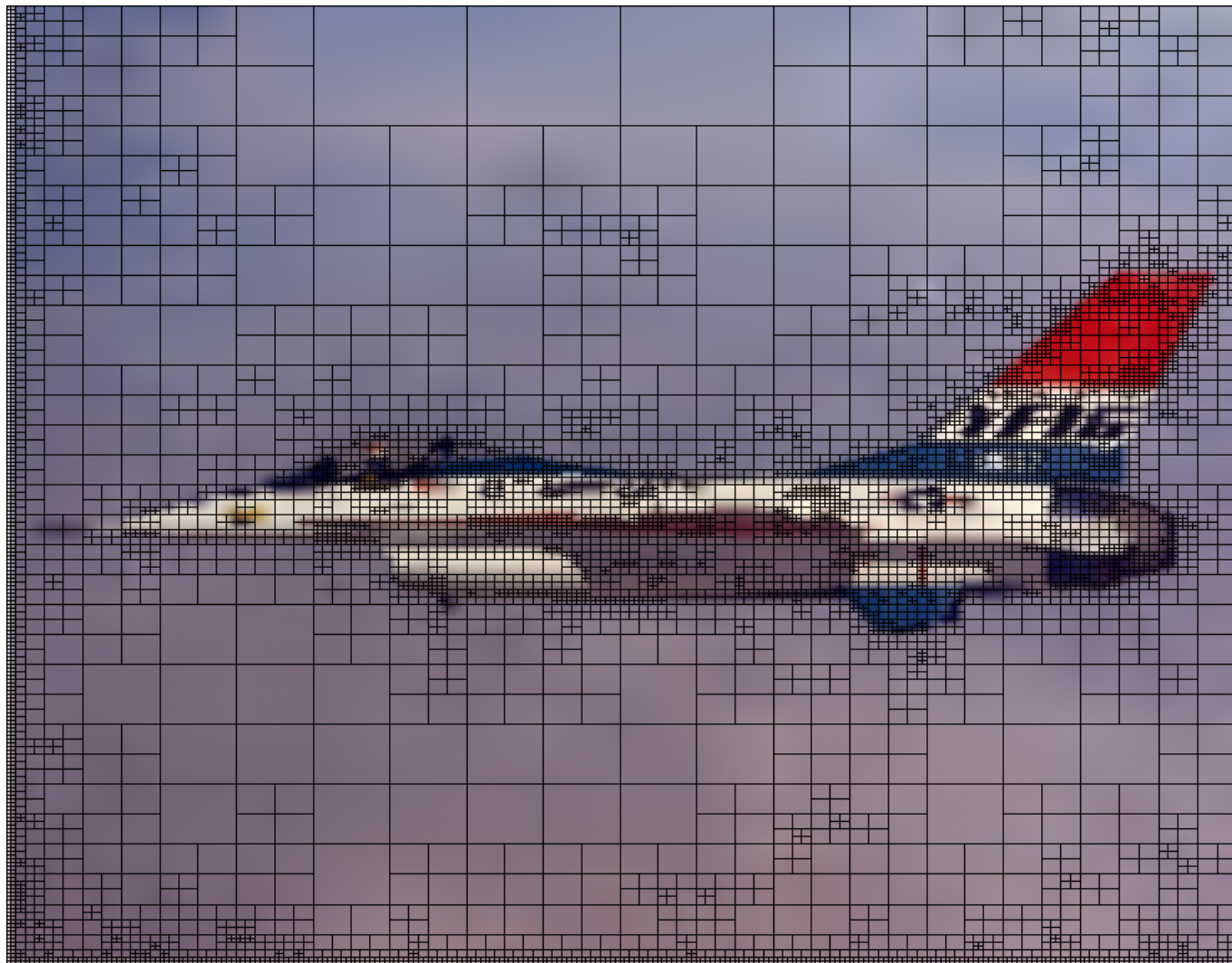
out/RGB_true_03_yf16.bmp



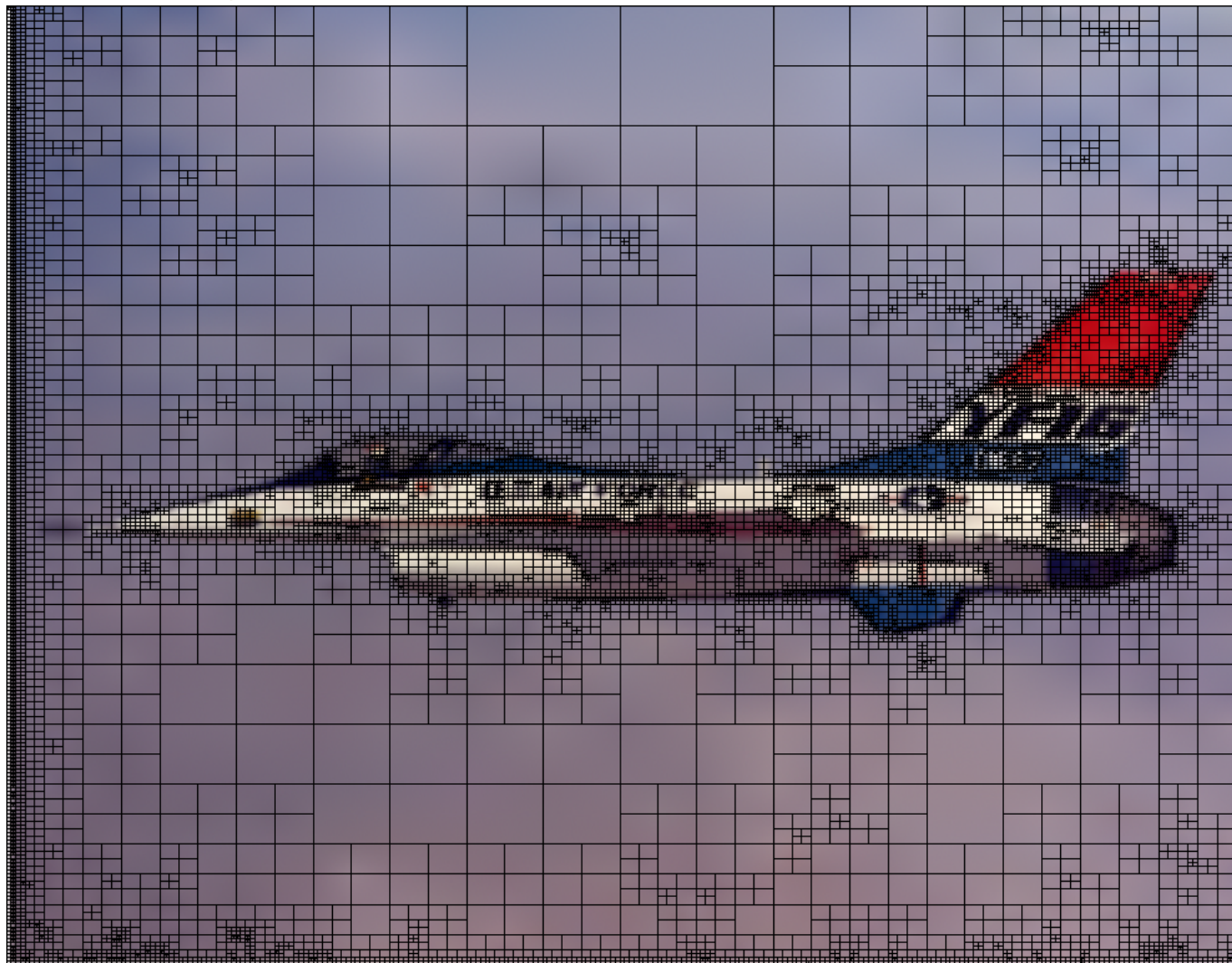
out/RGB_true_04_yf16.bmp



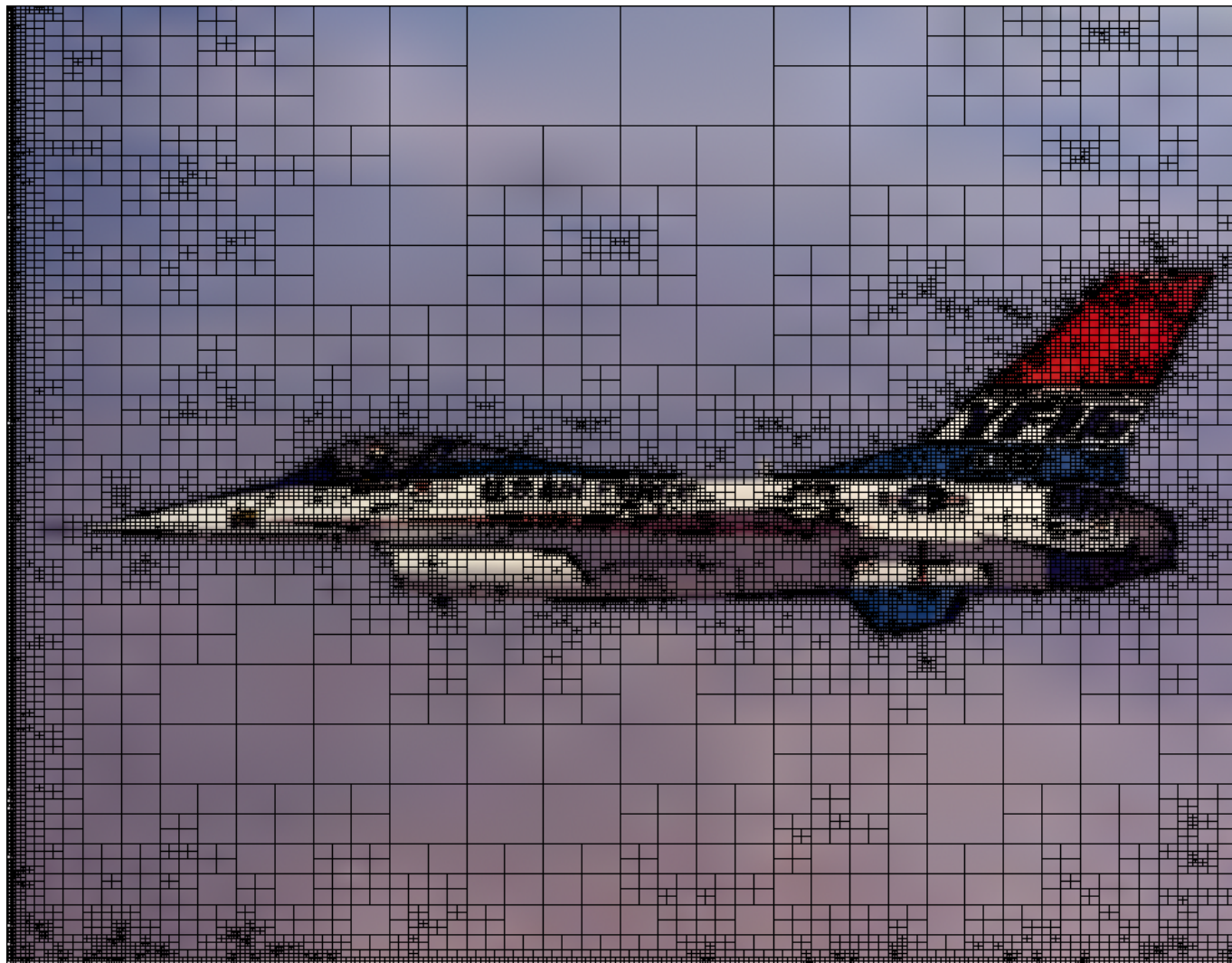
out/RGB_true_05_yf16.bmp



out/RGB_true_06_yf16.bmp



out/RGB_true_07_yf16.bmp



out/RGB_true_08_yf16.bmp

